Convention for citing the REACH regulation

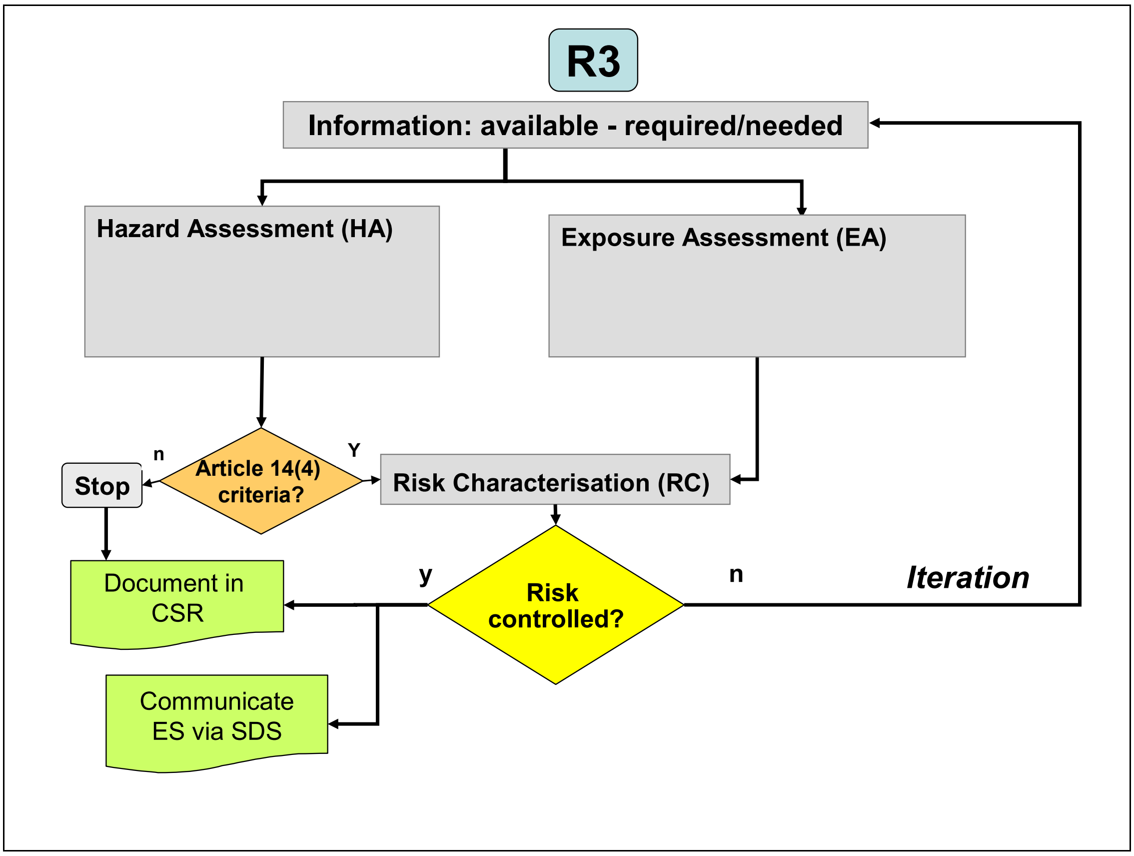
Where the REACH Regulation is cited literally, this is indicated by text in italics between quotes.

Table of Terms and Abbreviations

See Chapter R.20

Pathfinder

The figure below indicates the scope of part R.3 within the Guidance Document



R.3. INFORMATION GATHERING

This Guidance Document has been developed to assist registrants meet the information requirements for their chemicals by considering all types and sources of information and the adequacy and suitability of such data through specific Integrated Testing Strategies (ITS) for each endpoint.

However, before any of these strategies are applied, a critical first step is to assemble all of the available information on a substance, or information that may be useful to inform on the properties of that substance. This information should be used to drive the information gathering strategy detailed and described in subsequent specific chapters and is a vital first step in the overall process. This is envisaged in Step 1 of Annex VI of REACH - *Gather and share existing information* – and is described in Sub-section R.3.1.

The specific information requirements for REACH are detailed in Annexes VI-X of the Regulation and are discussed in detail in the endpoint specific subsections of Chapter R.7.

R.3.1 Information sources/searching

This section addresses information searching strategies and sources of information, but not the use or quality of the information that this process may yield.

Within the context of the Regulation, information is required for the specific purposes of:

- Classification and Labelling
- Determination of PBT status
- Determination of vPvB status
- Chemical Safety Assessment and Report
- Determination of any need for risk management measures

The above measures provide for:

- Downstream communication of information on the hazards of substances
- Ensuring a high level of protection of human health and the environment, as well as the free circulation of substances on the internal market while enhancing competitiveness and innovation

Failure to collate all of the available information on a substance may lead to duplicate work, wasted time, increased costs and potentially unnecessary animal use. Consequently, a thorough and reliable information gathering stage is a critical first step.

REACH requires the submission of data on:

- Substance identity
- Physico-chemical properties
- Exposure/uses/occurrence and applications
- Mammalian toxicity
- Environmental toxicity

- Environmental fate, including chemical and biotic degradation.

For the description of substance identity see the Guidance for identification and naming of substances under REACH. In many cases the information gathered may consist of actual test data. However, other types of information may be sufficient, especially when used in a *Weight of Evidence* approach. Such information could include:

- Human data
- Data from *in vivo* or *in vitro* studies that have not been generated in accordance with the latest adopted/accepted version of the corresponding (validated) test method or to GLP (or equivalent) standards
- (Q)SAR model outputs[2]
- SAR model outputs, read-across and category approaches

Consequently the information searching strategy needs to be as wide as possible. Guidance is given below on information sources specific to each endpoint and the searcher needs to understand the range of potential sources of information, and their content, structure, design and format. Given the large numbers of available resources and venues, the time required to learn the details of each system can be extensive, leading many searchers to search only a few, familiar resources. However this restricted approach is unlikely to yield all available data/information.

Information source types that could be included in any search strategy comprise (but are not limited to):

- in house Company and trade association files – may include studies generated in-house, commissioned studies carried out by contract houses, information on type and experience in use, reports from downstream companies and customers, purchased reports from other companies, collections of published papers and reviews of published data, and Safety Data Sheets. This kind of information requires expertise to interpret it. For studies not in the public domain there is the requirement to demonstrate legal title to the information in order to protect intellectual property rights of the data owner.
- Databanks and databases of compiled data – the content depends upon the objectives of the hosts/providers (which may change with time). Databanks generally contain limited information from original sources, but usually give little insight into test information quality. Databases and databanks should be seen only as routes to the cited original sources and are often indicative of the amount of published literature on a substance. They usually cover many more chemicals than the product range of any company. Requires expertise in searching numerous systems and in interpreting information.
- Published literature – could include papers reporting original findings (primary papers), review papers, books, monographs, and reports of proceedings, meetings and conferences. Covers many more chemicals than the product range of any company. Requires expertise in both identifying and interpreting information.
- Internet – search engines allow identification of electronic versions of a diverse range of data sources. In addition, websites of various expert organizations and regulatory bodies contain useful information. To obtain the information, registry numbers, chemical names and possible synonyms will need to be used in the search strategy.
- (Q)SAR models – some of these are available without charge and others are a fee based service. These sources are described in the generic section on QSAR and in the specific

[2] Detailed guidance on how to gather non-testing data is provided in Section R.6.1.7

sections for each endpoint. Specialised expertise is required to run models and interpret results. See Section R.6.1 for further guidance on these models and their use.

An indicative list of major available databases and databanks is given in Section R.3.4.

There is a large literature on the subject of literature searching itself. For example see the special edition of the Journal of Toxicology which was devoted to the topic. It is not the intention of this subchapter to detail this area of expertise although a bibliography is included in Section R.3.4 for further information / reading.

The internet is now maturing as a source of information, but the searcher needs to be aware that a variety of sources needs to be checked rather than just a single source, in order to be sure that all relevant external information is retrieved. Many of the most useful external sources of information are fee based services accessed through a data base vendor or specialist service provider. Sources vary in many aspects, including quality, reliability, and accuracy, indexing policy, extent of peer review, the time-spans covered, numbers of chemicals addressed and the extent of detail. Experienced searchers will know which sources have been most useful to them in the past. In some cases, comparative evaluations have been carried out. For example, Wright (2001) has given an overview of selected fee and non-fee databases, along with experience of the quality of service desk assistance.

The data table/summary databases are a source of initial information but the individual databases typically cover only a relatively small number of chemicals and endpoints. Consequently they cannot always be relied upon to be comprehensive so they often need to be supplemented with other databases including bibliographic ones. One strategy is to use initially free web based sources of information to locate information sources, and gauge the amount of data available. If little or no information is found then the more sophisticated sources may be interrogated. Some of these databases are complex and require knowledge of chemistry and chemical nomenclature to get the best from the investment of time and resource required. This is especially true of substructure searches that may be employed to look for information on similar or complementary substances where the information may be extremely useful for SAR relationships or within categories of substances. Consequently it may often be most cost effective to use a specialist information service provider to access all relevant sources with a consistent strategy.

The OECD has developed a web site giving free public access to existing information on existing Chemicals (The Global Portal to Information on Chemical Substances). In a first phase of development, the Portal gives access to many existing assessment reports and datasets – see http://webnet3.oecd.org/eChemPortal/ and can at this stage be queried by CAS No and chemical names. The OECD is investigating the feasibility of the development of a second phase where different databases that share the same data structure would be linked to the portal and thereby allowing the users to query the Portal by both current simple but also very advanced and complex search facilities including search possibilities related to chemical structure and properties. The European Commission and the US-EPA databases for their national/regional chemicals programmes will be linked to this second version of the Portal in a pilot phase. There are plans to extend this to other countries at a later stage.

R.3.2 Recording the Search Strategy

The exact searching strategy for a particular substance will be dependent on that substance - a proprietary molecule is unlikely to have any information in the public domain whereas for some high production volume substances the available information may be found in comprehensive reviews obtainable from international organisations. Whatever strategy is employed, it is important to record what is done and when. This serves two purposes: as a check on the detail and thoroughness of the search and also, if a search is repeated on a regular basis, it records the time and extent of the previous search. The search strategy should be recorded and there is a specific chapter in IUCLID 5 to record the details of the search that resulted in the information provided in the registration dossier. The major elements to capture in this record include:

- Chemicals names and synonyms used for the search
- <u>In house Company and trade association files</u>
    - Details of the database(s) and coverage
    - Date of search
- <u>Databanks and databases of compiled data</u>
    - <u>Published literature</u>
        - <u>Databank / Database name(s),</u>
        - <u>Calendar Years covered by the database / databank</u>
        - <u>Date of search</u>
- <u>Internet</u>
    - Search engines used
    - Date of search
- Textbooks Consulted
- <u>Other Sources of information</u>
- <u>(Q)SAR models</u>
    - Name of model / Software version / Reference

It is not the purpose of the search strategy record to document the validity of any QSAR model used, this will be done, as necessary, in the specific endpoint description where the QSAR is used.

When a search is done to find analogous substances for the purposed of chemical category formation, or for establishment of Structure Activity Relationships, the approach should be documented in a similar way.

For employment of chemical categories and (Q)SAR models further guidance is given in Sections R.6.1 and R.6.2.

For many high production volume substances there is an extensive literature and they have often been the subject of extensive critical reviews and evaluation for their effect on both health and the environment. These reviews, from regulatory, academic and international organisations are peer reviewed and generally accepted by stakeholders. Where such reviews exist an exhaustive literature search would only reveal significant amounts of data that have already been assessed and information published after the review has been produced.

Such reviews include (illustrative, not exhaustive)

- EU Existing Substance Regulation Risk assessment
- OECD SIDS evaluations
- WHO International Programme on Chemical Safety; e.g. Environmental Health Criteria documents and Concise International Assessment Documents
- WHO International Agency on Cancer - Monographs

- ECETOC Joint assessment of Commodity Chemical reports (JACC)
- National documentation; e.g. UK HSE Documentation for setting occupational exposure standards. BUA reports, US Environmental Protection Agency reports, BG Chemie

Deviations from reviews under EU legislation should be justified (see REACH Annex I paragraph 0.5.).

It would serve little to add to the overall assessment of a substance by revisiting all of the primary information sources cited in such reviews. In such cases these reviews should form the basis of the information collection strategy and help in both the identification of key studies and *Weight of Evidence* approaches. However, attention must be given to establishing the quality of such substance reviews, for example it is expected that it will have undergone a quality assurance procedure such as a peer review process. Furthermore, it would be necessary to determine when the last complete literature assessment was conducted for the specific substance review in order to ensure that no significant information has been published since the review literature search was conducted. As with all information collection strategies, the decision made for selecting a review and any additional information needs to be documented. It will be necessary to consult the primary literature in order to confirm the study outcomes that drive both classification and the Chemical Safety Assessment – see General Decision Making Framework (GDMF) step 3 in Chapter R.2.

R.3.3 Published Articles on Searching for Health/Hazard Information

1. Doldi, LM; Bratengeyer, E
The web as a free source for scientific information: a comparison with fee-based databases
ONLINE INFORMATION REVIEW, 29 (4): 400-411 2005

2. Wexler, P
The US National Library of Medicine's Toxicology and Environmental Health Information Program
TOXICOLOGY, 198 (1-3): 161-168 MAY 20 2004

3. Voigt, K; Welzl, G
Chemical databases: an overview of selected databases and evaluation methods
ONLINE INFORMATION REVIEW, 26 (3): 172-192 2002

4. Wexler, P
Introduction to special issue (part II) on digital information and tools
TOXICOLOGY, 173 (1-2): 1-1 APR 25 2002

5. Russom, CL
Mining environmental toxicology information: web resources
TOXICOLOGY, 173 (1-2): 75-88 APR 25 2002

6. Patterson, J; Hakkinen, PJB; Wullenweber, AE
Human health risk assessment: selected internet and world wide web resources
TOXICOLOGY, 173 (1-2): 123-143 APR 25 2002

7. Guerbet, M; Guyodo, G
Efficiency of 22 online databases in the search for physico-chemical, toxicological and ecotoxicological information on chemicals
ANNALS OF OCCUPATIONAL HYGIENE, 46 (2): 261-268 MAR 2002

8. Hull, RN; Ferguson, GM; Glaser, JD; et al.
Risk assessment resources on the World-wide Web (WWW)
HUMAN AND ECOLOGICAL RISK ASSESSMENT, 8 (2): 443-457 FEB 2002

9. Wexler, P
Introduction to special issue on digital information and tools
TOXICOLOGY, 157 (1-2): 1-2 JAN 12 2001

10. Wexler, P
TOXNET: An evolving web resource for toxicology and environmental health information
TOXICOLOGY, 157 (1-2): 3-10 JAN 12 2001

11. Poore, LM; King, G; Stefanik, K
Toxicology information resources at the Environmental Protection Agency
TOXICOLOGY, 157 (1-2): 11-23 JAN 12 2001

12. Brinkhuis, RP
Toxicology information from US government agencies
TOXICOLOGY, 157 (1-2): 25-49 JAN 12 2001

13. Stoss, FW
Subnational sources of toxicology information: state, territorial, tribal, county, municipal, and community resources online
TOXICOLOGY, 157 (1-2): 51-65 JAN 12 2001

14. Wright, LL
Searching fee and non-fee toxicology information resources: an overview of selected databases
TOXICOLOGY, 157 (1-2): 89-110 JAN 12 2001

15. Anderson, CA; Copestake, PT; Robinson, L
A specialist toxicity database (TRACE) is more effective than its larger, commercially available counterparts
TOXICOLOGY, 151 (1-3): 37-43 OCT 26 2000

16. Gehanno, JF; Paris, C; Thirion, B; et al.
Assessment of bibliographic databases performance in information retrieval for occupational and environmental toxicology
OCCUPATIONAL AND ENVIRONMENTAL MEDICINE, 55 (8): 562-566 AUG 1998

17. Ludl, H; Schope, K; Mangelsdorf, I
Searching for information on toxicological data of chemical substances in selected bibliographic databases - Selection of essential databases for toxicological researches
CHEMOSPHERE, 32 (5): 867-880 MAR 1996

<u>3 special issues in journal Toxicology devoted to this topic:</u>

TOXICOLOGY, 157 (1-2): JAN 12 2001, Special Issue on Digital Information and Tools.

1. Introduction to special issue on digital information and tools • EDITORIAL, Pages 1-2, Philip Wexler
2. TOXNET: An evolving web resource for toxicology and environmental health information • ARTICLE. Pages 3-10, Philip Wexler
3. Toxicology information resources at the Environmental Protection Agency • ARTICLE, Pages 11-23, Linda Miller Poore, Geffry King and Karen Stefanik
4. Toxicology information from US government agencies • ARTICLE, Pages 25-49, Randall P. Brinkhuis
5. Subnational sources of toxicology information: state, territorial, tribal, county, municipal, and community resources online • ARTICLE, Pages 51-65, Frederick W. Stoss
6. Professional Toxicology Societies: Web Based Resources • ARTICLE, Pages 67-76, James P. Kehrer and Jon Mirsalis
7. Toxicology and environmental digital resources from and for citizen groups • ARTICLE, Pages 77-88, Peter Montague and Maria B. Pellerano
8. Searching fee and non-fee toxicology information resources: an overview of selected databases • ARTICLE, Pages 89-110, Larry L. Wright
9. The IOMC organisations: a source of chemical safety information • ARTICLE, Pages 111-119, Fatoumata Keita-Ouane, Linda Durkee, Emmert Clevenstine, Michael Ruse, Zoltan Csizer, Peter Kearns and Achim Halpaap
10. Using internet search engines and library catalogues to locate toxicology information • ARTICLE, Pages 121-139, Laura Dassler Wukovitz
11. Digital toxicology education tools: education, training, case studies, and tutorials • ARTICLE, Pages 141-152, Jonathan F. Sharpe, David L. Eaton and Craig B. Marcus
12. Online resources for news about toxicology and other environmental topics • ARTICLE, Pages 153-164, Jeffrey C. South

TOXICOLOGY, 173 (1-2): APR 25 2002, Special Issue (Part 2) on Digital Information and Tools.

1. Introduction to special issue (part II) on digital information and tools • EDITORIAL, Page 1, Philip Wexler

TOXICOLOGY, 190 (1-2): AUG 21 2003, Digital Information and Tools, Part 3 – Global Web Resources.

R.3.4 Indicative list of maj ble databases and databanks

R.3.4.1 No fee sources

Source		Description
European Chemicals Bureau (ECB) European Chemical Substances Information System (ESIS) http://esis.jrc.ec.europa.eu/		Provides information on chemicals, related to: EINECS (European Inventory of Existing Commercial chemical Substances), ELINCS (European List of Notified Chemical Substances), NLP (No-Longer Polymers), HPVCs (High Production Volume Chemicals) and LPVCs (Low Production Volume Chemicals), including EU Producers/Importers lists, C&L (Classification and Labelling), Risk and Safety Phrases, Danger etc..., IUCLID Chemical Data Sheets, IUCLID Export Files, OECD-IUCLID Export Files, EUSES Export Files, Priority Lists, Risk Assessment process and tracking system in relation to Council Regulation (EEC) 793/93 also known as Existing Substances Regulation (ESR).
US National Library of Medicine (NLM), Specialized Information Sources (SIS)		Provides access to many excellent databases, see individual descriptions below
http://sis.nlm.nih.gov/enviro.html	rnational imates for Risk t	Risk information for over 600 chemicals from authoritative groups worldwide
	ated Risk System	Hazard identification and dose-response assessments for over 500 chemicals
	ardous Data Bank	Comprehensive, peer-reviewed toxicological data for over 5,000 chemicals Excerpts from published literature on: Human Health Effects and Emergency Medicine Treatment Animal Toxicity Studies Ecotoxicology Studies Environmental Fate and Exposure Chemical and Physical Properties Chemical Safety and Handling Environmental and Occupational Standards and Regulations Manufacturing and Use Information

Source		Description
	Toxicology nline	Over three million references from the toxicology literature, including MEDLINE/ PubMed, research in progress, and meeting abstracts
	- Chemical n Plus	Dictionary of over 380,000 chemicals (names, synonyms, structures). Includes links to NLM databases and other resources such as ATSDR Medical Management Guidelines and Public Health Statements
	- advanced	Provides structure search and display for over 260,000 chemicals Includes links to NLM databases and other resources
		Lists of chemicals of interest to government agencies
	emical esis Research System	Carcinogenicity, mutagenicity, tumour promotion, and tumour inhibition test results for over 9,000 chemicals
	elopmental and e Toxicology	Over 200,000 references to teratology, developmental and reproductive toxicology literature
	- Genetic Data Bank	Peer-reviewed genetic toxicology test data for over 3,000 chemicals
		Links jobs and hazardous tasks with occupational diseases and their symptoms
		Database of drugs to which breastfeeding mothers may be exposed. Covers maternal and infant drug levels possible effects on infants alternate drugs to consider
	Products	Potential health effects of chemicals for over 6000 common household products. Information in the Household Products Database is taken from a variety of publicly available sources, including brand-specific labels and Material Safety Data Sheets (MSDS) prepared by manufacturers
	Release	Annual environmental releases on over 600 toxic chemicals by U.S. facilities
		Geographic representation of TRI (US chemical releases) data with links to other TOXNET resources
http://sis.nlm.nih.gov/enviro/toxweblinks.html	Web Links – ources of data	

R.3.4.2 Fee based sources

Sources of Health and Environmental Ha ion

Databases	**Av**		**File Type**	**Subjects Covered**	**Years Included**
Agricola	Co	base vendors	Bibliographic, indexed	Agriculture, pesticides, human and environmental health	1970 - present
AMA Journals	Co	base vendors	Full text	Medicine, occupational medicine	1982 - present
Encompass Literature (previously APILIT – American Petroleum Institute)	Su ven	Commercial database sion	Bibliographic, extensive indexing, CAS RNs	Toxicology, environmental health, risk assessment	1963 - present
Aquaculture	Co	base vendors	Bibliographic, indexed	Environmental, aquatic toxicology	1970 - present
Aquatic Sciences & Fisheries Abstract	Co	base vendors	Bibliographic, indexed	Environmental, aquatic toxicology	1978 - present
Biological Abstracts – BIOSIS	Co	base vendors	Bibliographic, extensive indexing, CAS RNs	All aspects of biology including mammalian, human and environmental toxicology	1969 - present
CAB Abstracts	Co	base vendors	Bibliographic, indexed	Agriculture, pesticides, human and environmental health	1972 - present
Cancerlit	Co	base vendors	Bibliographic, extensive indexing, CAS RNs	Primarily human and animal chronic toxicology	1975 - 2002
Chemical Abstracts	Co	base vendors	Bibliographic, extensive indexing, CAS RNs	Mammalian, human and environmental toxicology, risk assessment	1967 - present
Chemical Abstracts Registry File	Co	base vendors	Extensive indexing, original source of CAS RNs	Index of all chemical compounds appearing in the published literature, includes physical/ chemical properties and indicators of amount of literature available	1967 - present
Chemical Carcinogenesis Research Info. Service – CCRIS	Co	base vendors	Data Tables/ Summaries	Cancer and chronic toxicity studies summarized	
Chemical Exposure	Co	base vendors	Bibliographic, indexed	Human exposures to chemicals and their health effects summarized, small	1974 - present

Databases	Av		File Type	Subjects Covered	Years Included
				database	
Chemical Information System (CIS) Databases: AQUIRE - Aquatic Information Retrieval CASR - Chemical Activity Status Report CESARS - Chemical Evaluation Search & Retrieval System ENVIROFATE - Environmental Fate GENETOX - Genetic Toxicity GIABS - Gastrointestinal Absorption ISHOW - Info. System for Hazardous Organics in Water OHM/TADS - Oil and Hazardous Materials/ Technical Assistance Data System PHYTOTOX - Terrestrial Plant Tox SANSS - Structure & Nomenclature Search System SUSPECT - Suspect Chemicals Source Book TSCATS - TSCA Submissions - Unpublished Data	Co	base vendors	Data Tables/ Summaries	Summarized results searchable by endpoint, species, and route of administration. Some very unique databases, such as PHYTOTOX which only covers effects on plants (primarily agriculture related)	Varies
Chemlist. Australian Inventory, status through June 1996 EINECS , DSL, NDSL status through June 15, 1990 EINECS PMNs (European List of Notified Chemical Substances - ELINCS) through March 2005 Japanese Existing and New Chemical Substances List (ENCS), status through Sept. 2004 Korean Existing Chemicals List (ECL) Inventory through December 2005 TSCA Actions, Inventory Status, and PMN's,	Co	base vendors	Indexed	Indication of hazard based on regulatory lists upon which the material appears, and provides a measure of how much published hazard information is likely to be found.	Varies

Databases	Av		File Type	Subjects Covered	Years Included
coverage through January 6, 2006					
Philippines Inventory of Chemicals and Chemical Substances status through 2004					
Swiss Inventory of Notified New Substances status through 2004					
Dissertation Abstracts	Co	base vendors	Bibliographic, indexed	All areas of health	1861 - present
EMBASE/Excerpta Medica	Co	base vendors	Bibliographic, extensive indexing, CAS RNs	Health and environmental related areas	1974 - present
Energy Science & Technology	Co	base vendors	Bibliographic, indexed	Primarily environmental effects	1974 – present
Engineering Index - Compendex	Co	base vendors	Bibliographic, extensive indexing, CAS RNs	Environmental engineering (air, water, pollution, solid waste)	1970 – present
Enviroline	Co	base vendors		Environmental effects (air, water, solid waste)	1970 – present
Environmental Bibliography	Co	base vendors		Environmental effects (air, water, solid waste)	1974 – present
EPA's Integrated Risk Information Service – IRIS	Co	base vendors	Data Tables/ Summaries	Summary of data used and cancer risk assessment done by the US-EPA	
ECB's ESIS – European Chemical Substances Information System	http	uropa.eu/	Data Tables/ Summaries	Summaries of data submitted to the EU (IUCLID, HPV data)	
GEOBASE	Co	base vendors	Bibliographic, indexed	Environmental effects (air, water, solid waste)	1980 – present
Hazardous Substances Data Bank – HSDB	Co	base vendors	Data Tables/ Summaries	Summaries of all health aspects including end use exposures/ measured levels in the ambient environment. Excellent database, peer reviewed but only covers a small number of chemicals.	
Life Sciences Collection	Co	base vendors	Bibliographic, indexed	All aspects of health/ hazard information.	1978 – present

Databases	Av		File Type	Subjects Covered	Years Included
JICST - EPlus (Japanese Science & Technology)	Co	base vendors	Bibliographic, indexed	Some coverage of health/hazard topics	1985 – present
Medline	Co	base vendors	Bibliographic, extensive indexing, CAS RNs	All aspects of health/ hazard information.	1960 – present
National Technical Information Service – NTIS	Co	base vendors	Bibliographic, indexed	All aspects of health/ hazard information published by US government.	1964 –present
NIOSH	Co	base vendors	Bibliographic, indexed	Occupational surveys and other related health information	1973 – 1998
Oceanic Abstracts	Co	base vendors	Bibliographic, indexed	Environmental effects	1964 – present
PASCAL	Co	base vendors	Bibliographic, indexed	All aspects of health/ hazard information focused on European publications	1973 – present
Pollution Abstracts	Co	base vendors	Bibliographic, indexed	Primarily environmental effects	1970 – present
Registry of Toxic Effects of Chemical Substances – RTECS	Co	base vendors	Data Tables/ Summaries	Toxicity, environmental data, lowest published toxicity values for each chemical listed	
Royal Society of Chemistry Databases: Chemical Hazards in Industry - CHI Laboratory Hazards Bulletin - LHB Chemical Safety NewsBase	Co	base vendors	Bibliographic, indexed	Toxicity, occupational hazards, exposures	 1984 - present 1981 - present 1981 – present
Science Citation Index	Co	base vendors	Bibliographic, indexed	Toxicology, environmental, risk assessments	1978 – present
TRACE	BIB	n Services Ltd	Bibliographic, indexed	Toxicology and Health effects of chemicals	1963 – present